

UNIT - 3

Sentiment Analysis

- 1. Sentiment analysis overview**
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Sentiment Analysis Overview

We, humans, are social beings. We are adept at utilizing a variety of means to communicate. We often consult financial discussion forums before making an investment decision; ask our friends for their opinions on a newly opened restaurant or a newly released movie; and conduct Internet searches and read consumer reviews and expert reports before making a big purchase like a house, a car, or an appliance. We rely on others' opinions to make better decisions, especially in an area

where we don't have a lot of knowledge or experience. Thanks to the growing availability and popularity of opinion-rich Internet resources such as social media outlets (e.g., Twitter, Facebook, etc.), online review sites, and personal blogs, it is now easier than ever to find opinions of others (thousands of them, as a matter of fact) on everything from the latest gadgets to political and public figures. Even though not everybody expresses opinions over the Internet, due mostly to the fastgrowing numbers and capabilities of social communication channels, the numbers are increasing exponentially.

Sentiment is a difficult word to define. It is often linked to or confused with other terms like belief, view, opinion, and conviction. Sentiment suggests a settled opinion reflective of one's feelings (Mejova, 2009). Sentiment has some unique

properties that set it apart from other concepts that we may want to identify in text. Often we want to categorize text by topic, which may involve dealing with whole taxonomies of topics.

Sentiment classification, on the other hand, usually deals with two classes (positive versus negative), a range of polarity (e.g., star ratings for movies), or even a range in strength of opinion (Pang and Lee, 2008). These classes span many topics, users, and documents. Although dealing with only a few classes may seem like an easier task than standard text analysis, it is far from the truth.

As a field of research, sentiment analysis is closely related to computational linguistics, natural language processing, and text mining. Sentiment analysis has many names. It's often referred to as opinion mining, subjectivity analysis, and appraisal extraction, with some connections to affective computing (computer recognition and expression of emotion). The sudden upsurge of interest and activity in the area of sentiment analysis(i.e., opinion mining), which deals with the automatic extraction of opinions, feelings, and subjectivity in text, is creating opportunities and threats for businesses and individuals alike. The ones who embrace and take advantage of it will greatly benefit from it.

Every opinion put on the Internet by an individual or a company will be accredited to the originator (good or bad) and will be retrieved and mined by others (often automatically by computer programs).

Sentiment analysis is trying to answer the question “What do people feel about a certain topic?” by digging into opinions of many using a variety of automated tools. Bringing together researchers and practitioners in business, computer science, computational linguistics, data mining, text mining, psychology, and even sociology, sentiment analysis aims to expand traditional fact-based text analysis to new frontiers, to realize opinion-oriented information systems.

In a business setting, especially in marketing and customer relationship management, sentiment analysis seeks to detect favorable and unfavorable opinions toward specific products and/or services using large numbers of textual data sources (customer feedback in the form of Web postings, tweets, blogs, etc.). Sentiment that appears in text comes in two flavors: explicit, where the subjective sentence directly expresses an opinion (“It’s a wonderful day”), and implicit, where the text implies an opinion (“The handle breaks too easily”). Most of the earlier work done in sentiment analysis focused on the first kind of sentiment, since it was easier to analyze.

Current trends are to implement analytical methods to consider both implicit and explicit sentiments. Sentiment polarity is a particular feature of text that sentiment analysis primarily focuses on. It is usually dichotomized into two—**positive** and **negative**—but polarity can also be thought of as a range. A document containing several opinionated statements would have a mixed polarity overall, which is different from not having a polarity at all (being objective) (Mejova, 2009).

Timely collection and analysis of textual data, which may be coming from a variety of sources—ranging from customer call center transcripts to social media postings—is a crucial part of the capabilities of proactive and customer-focused companies, nowadays.

Sentiment Analysis Applications

Compared to traditional sentiment analysis methods, which were survey based or focus group centered, costly, and time-consuming (and therefore driven from small samples of participants), the new face of text analytics—based sentiment analysis is a limit breaker. Current solutions automate very large-scale data collection, filtering, classification, and

clustering methods via natural language processing and data mining technologies that handle both factual and subjective information. Sentiment analysis is perhaps the most popular application of text analytics, tapping into data sources like tweets, Facebook posts, online communities, discussion boards, Web logs, product reviews, call center logs and recording, product rating sites, chat rooms, price comparison portals, search engine logs, and newsgroups. The following applications of sentiment analysis are meant to illustrate the power and the widespread coverage of this technology.

Voice of the customer (VOC)

Voice of the customer (VOC) is an integral part of an analytic CRM and customer experience management systems. As the enabler of VOC, sentiment analysis can access a company's product and service reviews (either continuously or periodically) to better understand and better manage the customer complaints and praises. For instance, a motion picture advertising/marketing company may detect the negative sentiments toward a movie that is about to open in theatres (based on its trailers), and quickly change the composition of trailers and advertising strategy (on all media outlets) to mitigate the negative impact. Similarly, a software company may detect the negative buzz regarding the bugs found in their newly released product early enough to release patches and quick fixes to alleviate the situation.

Often, the focus of VOC is individual customers, their service- and support-related needs, wants, and issues. VOC draw data from the full set of customer touch points, including e-mails, surveys, call center notes/recordings, and social media postings, and match customer voices to transactions (inquiries, purchases, returns) and individual customer profiles captured in enterprise operational systems. VOC, mostly driven by sentiment analysis, is a key element of customer experience management initiatives, where the goal is to create an intimate relationship with the customer.

Voice of the Market (VOM)

Voice of the market is about understanding aggregate opinions and trends. It's about knowing what stakeholders—customers, potential customers, influencers, whoever—are saying about your (and your competitors') products and services. A well-done VOM analysis helps companies with competitive intelligence and product development and positioning.

Voice of the Employee (VOE)

Traditionally VOE has been limited to employee satisfaction surveys. Text analytics in general (and sentiment analysis in particular) is a huge enabler of assessing the VOE. Using rich, opinionated textual data is an effective and efficient way

to listen to what employees are saying. As we all know, happy employees empower customer experience efforts and improve customer satisfaction.

Brand Management

Brand management focuses on listening to social media where anyone (past/current/prospective customers, industry experts, other authorities) can post opinions that can damage or boost your reputation. There are a number of relatively newly launched start-up companies that offer analytics-driven brand management services for others. Brand management is product and company (rather than customer) focused. It attempts to shape perceptions rather than to manage experiences using sentiment analysis techniques.

Financial Markets

Predicting the future values of individual (or a group of) stocks has been an interesting and seemingly unsolvable problem. What makes a stock (or a group of stocks) move up or down is anything but an exact science. Many believe that the stock market is mostly sentiment driven, making it anything but rational (especially for short-term stock movements). Therefore, use of sentiment analysis in financial markets has gained significant popularity. Automated analysis of market sentiments using social media, news,

blogs, and discussion groups seems to be a proper way to compute the market movements. If done correctly, sentiment analysis can identify short-term stock movements based on the buzz in the market, potentially impacting liquidity and trading.

Politics

As we all know, opinions matter a great deal in politics. Because political discussions are dominated by quotes, sarcasm, and complex references to persons, organizations, and ideas, politics is one of the most difficult, and potentially fruitful, areas for sentiment analysis. By analyzing the sentiment on election forums, one may predict who is more likely to win or lose. Sentiment analysis can help understand what voters are thinking and can clarify a candidate's position on issues. Sentiment analysis can help political organizations, campaigns, and news analysts to better understand which issues and positions matter the most to voters. The technology was successfully applied by both parties to the 2008 and 2012 American presidential election campaigns.

Government Intelligence

Government intelligence is another application that has been used by intelligence agencies. For example, it has been suggested that one could monitor sources for increases in hostile or negative communications. Sentiment analysis can allow the automatic analysis of the opinions that people submit about pending policy or government-regulation proposals. Furthermore, monitoring communications for spikes in negative sentiment may be of use to agencies like Homeland Security.

Other Interesting Areas

Sentiments of customers can be used to better design e-commerce sites (product suggestions, upsell/cross-sell advertising), better place advertisements (e.g., placing dynamic advertisement of products and services that consider the sentiment on the page the user is browsing), and manage opinion- or review-oriented search engines (i.e., an opinion-aggregation Web site, an alternative to sites like Epinions, summarizing user reviews). Sentiment analysis can help with e-mail filtration by categorizing and prioritizing incoming e-mails (e.g., it can detect strongly negative or flaming e-mails and forward them to the proper folder), as well as citation analysis, where it can determine whether an

author is citing a piece of work as supporting evidence or as research that he or she dismisses.

Sentiment analysis process

Because of the complexity of the problem (underlying concepts, expressions in text, context in which the text is expressed, etc.), there is no readily available standardized process to conduct sentiment analysis. However, based on the published work in the field of sensitivity analysis so far (both on research methods and range of applications), a multi-step, simple logical process, as given in below figure , seems to be an appropriate methodology for sentiment analysis. These logical steps are iterative (i.e., feedback, corrections, and iterations are part of the discovery process) and experimental in nature, and once completed and combined, capable of producing desired insight about the opinions in the text collection.

Step 1: Sentiment Detection

After the retrieval and preparation of the text documents, the first main task in sensitivity analysis is the detection of objectivity. Here the goal is to differentiate between a fact and

an opinion, which may be viewed as classification of text as objective or subjective. This may also be characterized as calculation of O-S Polarity (Objectivity-Subjectivity Polarity, which may be represented with a numerical value ranging from 0 to 1). If the objectivity value is close to 1, then there is no opinion to mine (i.e., it is a fact); therefore, the process goes back and grabs the next text data to analyze. Usually opinion detection is based on the examination of adjectives in text. For example, the polarity of “what a wonderful work” can be determined relatively easily by looking at the adjective.

Step 2: N-P Polarity Classification

The second main task is that of polarity classification. Given an opinionated piece of text, the goal is to classify the opinion as falling under one of two opposing sentiment polarities, or locate its position on the continuum between these two polarities (Pang and Lee, 2008). When viewed as a binary feature, polarity classification is the binary classification task of labeling an opinionated document as expressing either an overall positive or an overall negative opinion (e.g., thumbs up or thumbs down). In addition to the identification of N-P polarity, one should also be interested in identifying the strength of the sentiment (as opposed to just positive, it may be expressed as mildly, moderately, strongly, or very strongly positive). Most of this research was done on product or movie reviews where the definitions of “positive”

and “negative” are quite clear. Other tasks, such as classifying news as “good” or “bad,” present some difficulty. For instance an article may contain negative news without explicitly using any subjective words or terms. Furthermore, these classes usually appear intermixed when a document expresses both positive and negative sentiments. Then the task can be to identify the main (or dominating) sentiment of the document. Still, for lengthy texts, the tasks of classification may need to be done at several levels: term, phrase, sentence, and perhaps document level. For those, it is common to use the outputs of one level as the inputs for the next higher layer. Several methods used to identify the polarity and strengths of the polarity are explained in the next section.

Step 3: Target Identification

The goal of this step is to accurately identify the target of the expressed sentiment (e.g., a person, a product, an event, etc.). The difficulty of this task depends largely on the domain of the analysis. Even though it is usually easy to accurately identify the target for product or movie reviews, because the review is directly connected to the target, it may be quite challenging in other domains. For instance, lengthy, general-purpose text such as Web pages, news articles, and blogs do not always have a predefined topic that they are assigned to, and often mention many objects, any of which

may be deduced as the target. Sometimes there is more than one target in a sentiment sentence, which is the case in comparative texts. A subjective comparative sentence orders objects in order of preferences—for example, “This laptop computer is better than my desktop PC.” These sentences can be identified using comparative adjectives and adverbs (more, less, better, longer), superlative adjectives (most, least, best), and other words (such as same, differ, win, prefer, etc.). Once the sentences have been retrieved, the objects can be put in an order that is most representative of their merits, as described in text.

Step 4: Collection and Aggregation

Once the sentiments of all text data points in the document are identified and calculated, in this step they are aggregated and converted to a single sentiment measure for the whole document. This aggregation may be as simple as summing up the polarities and strengths of all texts, or as complex as using semantic aggregation techniques from natural language processing to come up with the ultimate sentiment.

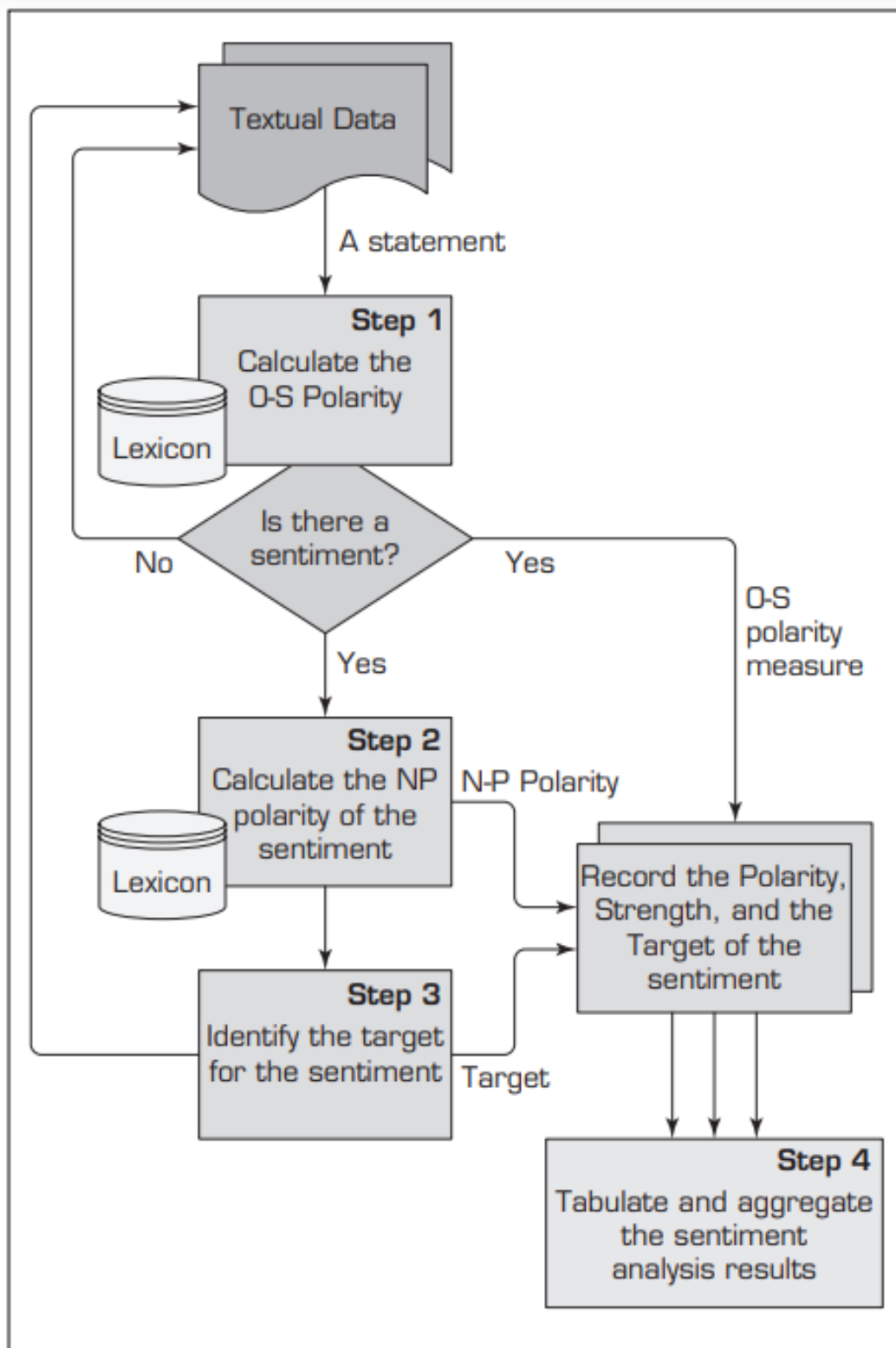


Figure: A Multi-Step Process to Sentiment Analysis.

Methods for Polarity Identification

As mentioned in the previous section, **polarity identification**—identifying the polarity of a text—can be made at the word, term, sentence, or document level. The most granular level for polarity identification is at the word level. Once the polarity identification is made at the word level, then it can be aggregated to the next higher level, and then the next until the level of aggregation desired from the sentiment analysis is reached. There seem to be two dominant techniques used for identification of polarity at the word/term level, each having its advantages and disadvantages:

1. Using a lexicon as a reference library (either developed manually or automatically, by an individual for a specific task or developed by an institution for general use)
2. Using a collection of training documents as the source of knowledge about the polarity of terms within a specific domain (i.e., inducing predictive models from opinionated textual documents)

Using a Lexicon

A lexicon is essentially the catalog of words, their synonyms, and their meanings for a given language. In addition to lexicons for many other languages, there are several general-purpose lexicons created for English. Often

general-purpose lexicons are used to create a variety of special-purpose lexicons for use in sentiment analysis projects. Perhaps the most popular general-purpose lexicon is WordNet, created at Princeton University, which has been extended and used by many researchers and practitioners for sentiment analysis purposes. As described on the WordNet Web site (**wordnet.princeton.edu**), it is a large lexical database of English, including nouns, verbs, adjectives, and adverbs grouped into sets of cognitive synonyms (i.e., synsets), each expressing a distinct concept. Synsets are interlinked by means of conceptual-semantic and lexical relations.

An interesting extension of WordNet was created by Esuli and Sebastiani (2006) where they added polarity (Positive-Negative) and objectivity (Subjective-Objective) labels for each term in the lexicon. To label each term, they classify the synset (a group of synonyms) to which this term belongs using a set of ternary classifiers (a measure that attaches to each object exactly one out of three labels), each of them capable of deciding whether a synset is Positive, or Negative, or Objective. The resulting scores range from 0.0 to 1.0, giving a graded evaluation of opinion-related properties of the terms. These can be summed up visually as in Below Figure . The edges of the triangle represent one of the three classifications (positive, negative, and objective). A term can be located in this space as a point, representing the extent to which it belongs to each of the classifications.

A similar extension methodology is used to create SentiWordNet, a publicly available lexicon specifically developed for opinion mining (sentiment analysis) purposes.

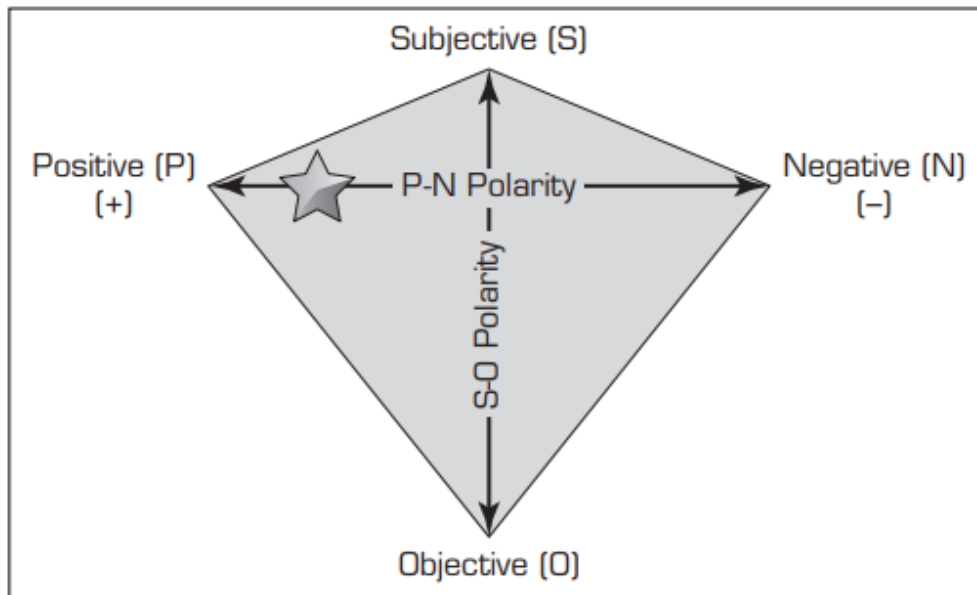


Fig: A Graphical Representation of the P-N Polarity and S-O Polarity Relationship.

SentiWordNet assigns to each synset of WordNet three sentiment scores: positivity, negativity, objectivity. More about SentiWordNet can be found at **sentiwordnet.isti.cnr.it**.

Another extension to WordNet is WordNet-Affect, developed by Strapparava and Valitutti (Strapparava and Valitutti, 2004). They label WordNet synsets using affective labels representing different affective categories like emotion, cognitive state, attitude, feeling, and so on. WordNet has also

been directly used in sentiment analysis. For example, Kim and Hovy (Kim and Hovy, 2004) and Hu and Liu (Hu and Liu, 2005) generate lexicons of positive and negative terms by starting with a small list of “seed” terms of known polarities (e.g., love, like, nice, etc.) and then using the antonymy and synonymy properties of terms to group them into either of the polarity categories.

Using a Collection of Training Documents

It is possible to perform sentiment classification using statistical analysis and machinelearning tools that take advantage of the vast resources of labeled (manually by annotators or using a star/point system) documents available. Product review Web sites like Amazon, C-NET, ebay, RottenTomatoes, and the Internet Movie Database (IMDB) have all been extensively used as sources of annotated data. The star (or tomato, as it were) system provides an explicit label of the overall polarity of the review, and it is often taken as a gold standard in algorithm evaluation.

A variety of manually labeled textual data is available through evaluation efforts such as the Text REtrieval Conference (TREC), NII Test Collection for IR Systems (NTCIR), and Cross Language Evaluation Forum (CLEF). The data sets these efforts produce often serve as a standard in the text mining community, including for

sentiment analysis researchers. Individual researchers and research groups have also produced many interesting data sets. Technology Insights lists some of the most popular ones. Once an already labeled textual data set is obtained, a variety of predictive modeling and other machine-learning algorithms can be used to train sentiment classifiers. Some of the most popular algorithms used for this task include artificial neural networks, support vector machines, k-nearest neighbor, Naive Bayes, decision trees, and expectation maximization-based clustering.

Identifying Semantic Orientation of Sentences and Phrases

Once the semantic orientation of individual words has been determined, it is often desirable to extend this to the phrase or sentence the word appears in. The simplest way to accomplish such aggregation is to use some type of averaging for the polarities of words in the phrases or sentences. Though rarely applied, such aggregation can be as complex as using one or more machine-learning techniques to create a predictive relationship between the words (and their polarity values) and phrases or sentences.

Identifying Semantic Orientation of Document

Even though the vast majority of the work in this area is done in determining semantic orientation of words and phrases/sentences, some tasks like summarization and information retrieval may require semantic labeling of the whole document (REF). Similar to the case in aggregating sentiment polarity from word level to phrase or sentence level, aggregation to document level is also accomplished by some type of averaging. Sentiment orientation of the document may not make sense for very large documents; therefore, it is often used on small to medium-sized documents posted on the Internet.

Technology Insights Large Textual Data Sets for Predictive Text Mining and Sentiment Analysis

Congressional Floor-Debate Transcripts: Published by Thomas et al. (Thomas and B. Pang, 2006); contains political speeches that are labeled to indicate whether the speaker supported or opposed the legislation discussed.

Econominig: Published by Stern School at New York University; consists of feedback postings for merchants at Amazon.com.

Cornell Movie-Review Data Sets: Introduced by Pang and Lee (Pang and Lee, 2008); contains 1,000 positive and 1,000 negative automatically derived document-level labels, and 5,331 positive and 5,331 negative sentences/snippets.

Stanford—Large Movie Review Data Set: A set of 25,000 highly polar movie reviews for training, and 25,000 for testing. There is additional unlabeled data for use as well. Raw text and already processed bag-of-words formats are provided. (See: [http:// ai.stanford.edu/~amaas/data/sentiment](http://ai.stanford.edu/~amaas/data/sentiment).)

MPQA Corpus: Corpus and Opinion Recognition System corpus; contains 535 manually annotated news articles from a variety of news sources containing labels for opinions and private states (beliefs, emotions, speculations, etc.).

Multiple-Aspect Restaurant Reviews: Introduced by Snyder and Barzilay (Snyder and Barzilay, 2007); contains 4,488 reviews with an explicit 1-to-5 rating for five different aspects: food, ambiance, service, value, and overall experience.

Sentiment Analysis and Speech Analytics

Speech analytics is a growing field of science that allows users to analyze and extract information from both live and recorded conversations. It is being used effectively to gather intelligence for security purposes, to enhance the presentation and utility of rich media applications, and perhaps most significantly, to deliver meaningful and quantitative business intelligence through the analysis of the millions of recorded calls that occur in customer contact centers around the world.

Sentiment analysis, as it applies to speech analytics, focuses specifically on assessing the emotional states expressed in a conversation and on measuring the presence and strength of positive and negative feelings that are exhibited by the participants. One common use of sentiment analysis within contact centers is to provide insight into a customer's feelings about an organization, its products, services, and customer service processes, as well as an individual agent's behavior. Sentiment analysis data can be used across an organization to aid in customer relationship management, agent training, and in identifying and resolving troubling issues as they emerge.

How Is It Done?

The core of automated sentiment analysis centers around creating a model to describe how certain features and

content in the audio relate to the sentiments being felt and expressed by the participants in the conversation. Two primary methods have been deployed to predict sentiment within audio: acoustic/phonetic and linguistic modeling.

The Acoustic Approach

The acoustic approach to sentiment analysis relies on extracting and measuring a specific set of features (e.g., tone of voice, pitch or volume, intensity and rate of speech) of the audio. These features can in some circumstances provide basic indicators of sentiment. For example, the speech of a surprised speaker tends to become somewhat faster, louder, and higher in pitch. Sadness and depression are presented as slower, softer, and lower in pitch (see Moore et al., 2008). An angry caller may speak much faster, much louder, and will increase the pitch of stressed vowels. There is a wide variety of audio features that can be measured.

The most common ones are as follows:

- Intensity: energy, sound pressure level
- Pitch: variation of fundamental frequency
- Jitter: variation in amplitude of vocal fold movements
- Shimmer: variation in frequency of vocal fold movements
- Glottal pulse: glottal-source spectral characteristics
- HNR: harmonics-to-noise ratio

- Speaking rate: number of phonemes, vowels, syllables, or words per unit of time

When developing an acoustic analysis tool, the system must be built on a model that defines the sentiments being measured. The model is based on a database of the audio features (some of which are listed here) and how their presence may indicate each of the sentiments (as simple as positive, negative, neutral, or refined, such as fear, anger, sadness, hurt, surprise, relief, etc.) that are being measured. To create this database, each single-emotion example is preselected from an original set of recordings, manually reviewed, and annotated to identify which sentiment it represents. The final acoustic analysis tools are then trained (using data mining techniques) and a predictive model is tested and validated using a different set of the same annotated recordings.

As sophisticated as it sounds, the acoustic approach has its deficiencies. First, because acoustic analysis relies on identifying the audio characteristics of a call, the quality of the audio can significantly impact the ability to identify these features. Second, speakers often express blended emotions, such as both empathy and annoyance (as in “I do understand, madam, but I have no miracle solution”), which are extremely difficult to classify based solely on their acoustic features. Third, acoustic analysis is often incapable

of recognizing and adjusting for the variety of ways that different callers may express the same sentiment. Finally, its time-demanding and laborious process make it impractical for use with live audio streams.

The Linguistic Approach

Conversely, the linguistic approach focuses on the explicit indications of sentiment and context of the spoken content within the audio; linguistic models acknowledge that, when in a charged state, the speaker has a higher probability of using specific words, exclamations, or phrases in a particular order.

The features that are most often analyzed in a linguistic model include:

- Lexical: words, phrases, and other linguistic patterns
- Disfluencies: filled pauses, hesitation, restarts, and nonverbals such as laughter or breathing
- Higher semantics: taxonomy/ontology, dialogue history, and pragmatics

The simplest method, in the linguistic approach, is to catch within the audio a limited number of specific keywords (a specific lexicon) that has domain-specific sentiment significance. This approach is perhaps the least popular due

to its limited applicability and less-than-desired prediction accuracy. Alternatively, as with the acoustic approach, a model is built based on understanding which linguistic elements are predictors of particular sentiments, and this model is then run against a series of recordings to determine the sentiments that are contained therein. The challenge with this approach is in collecting the linguistic information contained in any corpus of audio. This has traditionally been done using a large vocabulary continuous speech recognition (LVCSR) system, often referred to as speech-to-text. However, LVCSR systems are prone to creating significant error in the textual indexes they create. In addition, the level of computational effort they require—that is, the amount of computer processing power needed to analyze large amounts of audio content—has made them very expensive to deploy for mass audio analysis.

Yet, another approach to linguistic analysis is that of phonetic indexing and search. Among the significant advantages associated with this approach to linguistic modeling is the method's ability to maintain a high degree of accuracy no matter what the quality of the audio source, and its incorporation of conversational context through the use of structured queries during analysis (Nexidia, 2009).

UNIT - 4

Web Analytics and Web Mining

1. Security First Insurance
Deepens Connection with
Policyholders
2. Web Mining Overview
3. Web Content and Web Structure
Mining
4. Search Engines
5. Search Engine Optimization
6. Web Usage Mining (Web
Analytics)
7. Web Analytics Maturity Model
and Web Analytics Tools

Security First Insurance Deepens Connection with Policyholders

Security First Insurance is one of the largest homeowners' insurance companies in Florida. Headquartered in Ormond Beach, it employs more than 80 insurance professionals to serve its nearly 190,000 customers.

CHALLENGE

Being There for Customers Storm After Storm, Year After Year

Florida has more property and people exposed to hurricanes than any state in the country. Each year, the Atlantic Ocean averages 12 named storms and nine named hurricanes. Security First is one of a few Florida homeowners' insurance companies that has the financial strength to withstand multiple natural disasters. "One of our promises is to be there for our customers, storm after storm, year after year," says Werner Kruck, chief operating officer for Security First.

During a typical month, Security First processes 700 claims. However, in the aftermath of a hurricane, that number can swell to tens of thousands within days. It can be a

challenge for the company to quickly scale up to handle the influx of customers trying to file post-storm insurance claims for damaged property and possessions. In the past, customers submitted claims primarily by phone and sometimes email. Today, policyholders use any means available to connect with an agent or claims representative, including posting a question or comment on the company's Facebook page or Twitter account.

Although Security First provides ongoing monitoring of its Facebook and Twitter accounts, as well as its multiple email addresses and call centers, the company knew that the communication volume after a major storm required a more aggressive approach. "We were concerned that if a massive number of customers contacted us through email or social media after a hurricane, we would be unable to respond quickly and appropriately," Kruck says. "We need to be available to our customers in whatever way they want to contact us." In addition, Security First recognized the need to integrate its social media responses into the claims process and document those responses to comply with industry regulations.

SOLUTION

Providing Responsive Service No Matter How Customers Get in Touch

Security First contacted IBM Business Partner Integritie for help with harnessing social media to improve the customer experience. Integritie configured a solution built on key IBM Enterprise Content Management software components, featuring IBM Content Analytics with Enterprise Search, IBM Content Collector for Email and IBM® FileNet® Content Manager software. Called Social Media Capture (SMC4), the Integritie solution offers four critical capabilities for managing social media platforms: capture, control, compliance and communication. For example, the SMC4 solution logs all social networking interaction for Security First, captures content, monitors incoming and outgoing messages and archives all communication for compliance review.

Because the solution uses open IBM Enterprise Content Management software, Security First can easily link it to critical company applications, databases and processes.

For example, Content Collector for Email software automatically captures email content and attachments and sends an email back to the policyholder acknowledging receipt. In addition, Content Analytics with Enterprise Search software sifts through and analyzes the content of customers'

posts and emails. The software then captures information gleaned from this analysis directly into claims documents to begin the claims process. Virtually all incoming communication from the company's web, the Internet and emails is pulled into a central FileNet Content Manager software repository to maintain, control and link to the appropriate workflow. "We can bring the customer conversation and any pictures and attachments into our policy and claims management system and use it to trigger our claims process and add to our documentation," says Kruck.

Prioritizing Communications with Access to Smarter Content

People whose homes have been damaged or destroyed by a hurricane are often displaced quickly, with little more than the clothes on their backs. Grabbing an insurance policy on the way out the door is often an afterthought. They're relying on their insurance companies to have the information they need to help them get their lives back in order as quickly as possible. When tens of thousands of policyholders require assistance within a short period of time, Security First must triage requests quickly. The Content Analytics with Enterprise Search software that anchors the SMC4 solution provides the information necessary to help the company identify and address the most urgent cases first. The software

automatically sifts through data in email and social media posts, tweets and comments using text mining, text analytics, natural language processing and sentiment analytics to detect words and tones that identify significant property damage or that convey distress. Security First can then prioritize the messages and route them to the proper personnel to provide reassurance, handle complaints or process a claim. “With access to smarter content, we can respond to our customers in a more rapid, efficient and personalized way,” says Kruck. “When customers are having a bad experience, it’s really important to get to them quickly with the level of assistance appropriate to their particular situations.”

RESULTS

Successfully Addressing Potential Compliance Issues

Companies in all industries must stay compliant with new and emerging regulatory requirements regarding social media. The text analysis capabilities provided in the IBM software help Security First filter inappropriate incoming communications and audit outbound communications, avoiding potential issues with message content. The company can be confident that the responses its employees provide are compliant and controlled based on both Security First policies and industry regulations.

Security First can designate people or roles in the organization that are authorized to create and submit responses. The system automatically verifies these designations and analyzes outgoing message content, stopping any ineffective or questionable communications for further review. “Everything is recorded for compliance, so we can effectively track and maintain the process. We have the ability to control which employees respond, their level of authority and the content of their responses,” says Kruck.

These capabilities give Security First the confidence to expand its use of social media. Because compliance is covered, the company can focus on additional opportunities for direct dialog with customers. Before this solution, Security First filtered customer communications through agents. Now it can reach out to customers directly and proactively as a company.

“We’re one of the first insurance companies in Florida to make ourselves available to customers whenever, wherever and however they choose to communicate. We’re also managing internal processes more effectively and proactively, reaching out to customers in a controlled and compliant manner,” says Kruck.

Some of the prevailing business benefits of creative use of Web and social analytics include:

- Turns social media into an actionable communications channel during a major disaster

- Speeds claims processes by initiating claims with information from email and social media posts
 - Facilitates prioritizing urgent cases by analyzing social media content for sentiments
- Helps ensure compliance by automatically documenting social media communications

Key learning by Security First Insurance Deepens Connection with Policyholders :

Web analytics is becoming a way of life for many businesses, especially the ones that are directly facing the consumers. Companies are expected to find new and innovative ways to connect with their customers, understand their needs, wants, and opinions, and proactively develop products and services that fit well with them. In this day and age, asking customers to tell you exactly what they like and dislike is not a viable option. Instead, businesses are expected to deduce that information by applying advanced analytics tools to invaluable data generated on the Internet and social media sites (along with corporate databases). Security First realized the need to revolutionize their business processes to be more effective and efficient in the way that they deal with their customers and customer claims. They not only used what the Internet and social media have to offer,

but also tapped into the customer call records/recordings and other relevant transaction databases. This vignette illustrates the fact that analytics technologies are advanced enough to bring together many different data sources to create a holistic view of the customer. And that is perhaps the greatest success criterion for today's businesses.

Web Mining Overview

The Internet has forever changed the landscape of business as we know it. Because of the highly connected, flattened world and broadened competitive field, today's companies are increasingly facing greater opportunities (being able to reach customers and markets that they may have never thought possible) and bigger challenge (a globalized and ever-changing competitive marketplace). Ones with the vision and capabilities to deal with such a volatile environment are greatly benefiting from it, while others who resist are having a hard time surviving. Having an engaged presence on the Internet is not a choice anymore: It is a business requirement. Customers are expecting companies to offer their products and/or services over the Internet. They are not only buying products and services but also talking about companies and sharing their transactional and usage experiences with others over the Internet.

The growth of the Internet and its enabling technologies has made data creation, data collection, and data/information/opinion exchange easier. Delays in service, manufacturing, shipping, delivery, and customer inquiries are no longer private incidents and are accepted as necessary evils. Now, thanks to social media tools and technologies on the Internet, everybody knows everything. Successful companies are the ones who embrace these Internet technologies and use them for the betterment of their business processes so that they can better communicate with their customers, understanding their needs and wants and serving them thoroughly and expeditiously. Being customer focused and keeping customers happy have never been as important a concept for businesses as they are now, in this age of the Internet and social media.

The World Wide Web (or, for short, the Web) serves as an enormous repository of data and information on virtually everything one can conceive—business, personal, you name it; an abundant amount of it is there. The Web is perhaps the world's largest data and text repository, and the amount of information on the Web is growing rapidly. A lot of interesting information can be found online: whose homepage is linked to which other pages, how many people have links to a specific Web page, and how a particular site is organized. In addition, each visitor to a Web site, each search on a search engine, each click on a link, and each transaction on an e-commerce site create additional data. Although

unstructured textual data in the form of Web pages coded in HTML or XML is the dominant content of the Web, the Web infrastructure also contains hyperlink information (connections to other Web pages) and usage information (logs of visitors' interactions with Web sites), all of which provide rich data for knowledge discovery. Analysis of this information can help us make better use of Web sites and also aid us in enhancing relationships and value for the visitors to our own Web sites.

Because of its sheer size and complexity, mining the Web is not an easy undertaking by any means. The Web also poses great challenges for effective and efficient knowledge discovery (Han and Kamber, 2006):

- **The Web is too big for effective data mining:**

The Web is so large and growing so rapidly that it is difficult to even quantify its size. Because of the sheer size of the Web, it is not feasible to set up a data warehouse to replicate, store, and integrate all of the data on the Web, making data collection and integration a challenge.

- **The Web is too complex**

The complexity of a Web page is far greater than a page in a traditional text document collection. Web pages lack a unified structure. They contain far more authoring style and content variation than any set of books, articles, or other traditional text-based document.

- **The Web is too dynamic:**

The Web is a highly dynamic information source. Not only does the Web grow rapidly, but its content is constantly being updated. Blogs, news stories, stock market results, weather reports, sports scores, prices, company advertisements, and numerous other types of information are updated regularly on the Web.

- **The Web is not specific to a domain**

The Web serves a broad diversity of communities and connects billions of workstations. Web users have very different backgrounds, interests, and usage purposes. Most users may not have good knowledge of the structure of the information network and may not be aware of the heavy cost of a particular search that they perform.

- **The Web has everything**

Only a small portion of the information on the Web is truly relevant or useful to someone (or some task). It is said that 99 percent of the information on the Web is useless to 99 percent of Web users. Although this may not seem obvious, it is true that a particular person is generally interested in only a tiny portion of the Web, whereas the rest of the Web contains information that is uninteresting to the user and may swamp desired results. Finding the portion of the Web that is truly relevant to a person and the task being performed is a prominent issue in Web-related research.

These challenges have prompted many research efforts to enhance the effectiveness and efficiency of discovering and using data assets on the Web. A number of index-based Web search engines constantly search the Web and index Web pages under certain keywords. Using these search engines, an experienced user may be able to locate documents by providing a set of tightly constrained keywords or phrases. However, a simple keyword-based search engine suffers from several deficiencies. First, a topic of any breadth can easily contain hundreds or thousands of documents. This can lead to a large number of document entries returned by the search engine, many of which are marginally relevant to the topic. Second, many documents that are highly relevant to a topic may not contain the exact keywords defining them. As we will cover in more detail later in this chapter, compared to keyword-based Web search, Web mining is a prominent (and more challenging) approach that can be used to substantially enhance the power of Web search engines because Web mining can identify authoritative Web pages, classify Web documents, and resolve many ambiguities and subtleties raised in keyword-based Web search engines.

Web mining (or Web data mining) is the process of discovering intrinsic relationships (i.e., interesting and useful information) from Web data, which are expressed in the form of textual, linkage, or usage information. The term Web mining was first used by Etzioni (1996); today, many

conferences, journals, and books focus on Web data mining. It is a continually evolving area of technology and business practice. Web mining is essentially the same as data mining that uses data generated over the Web. The goal is to turn vast repositories of business transactions, customer interactions, and Web site usage data into actionable information (i.e., knowledge) to promote better decision making throughout the enterprise. Because of the increased popularity of the term analytics, nowadays many have started to call Web mining Web analytics. However, these two terms are not the same. Although Web analytics is primarily Web site usage data focused, Web mining is inclusive of all data generated via the Internet, including transaction, social, and usage data. While Web analytics aims to describe what has happened on the Web site (employing a predefined, metrics-driven descriptive analytics methodology), Web mining aims to discover previously unknown patterns and relationships (employing a novel predictive or prescriptive analytics methodology). From a big-picture perspective, Web analytics can be considered a part of Web mining. Figure 8.1 presents a simple taxonomy of Web mining, where it is divided into three main areas: Web content mining, Web structure mining, and Web usage mining. In the figure, the data sources used in these three main areas are also specified. Although these three areas are shown separately, as you will see in the following section, they are often used

collectively and synergistically to address business problems and opportunities.

Below Figure indicates, Web mining relies heavily on data mining and text mining and their enabling tools and techniques, which we have covered in detail in the previous two units . The figure also indicates that these three generic areas are further extended into several very well-known application areas.

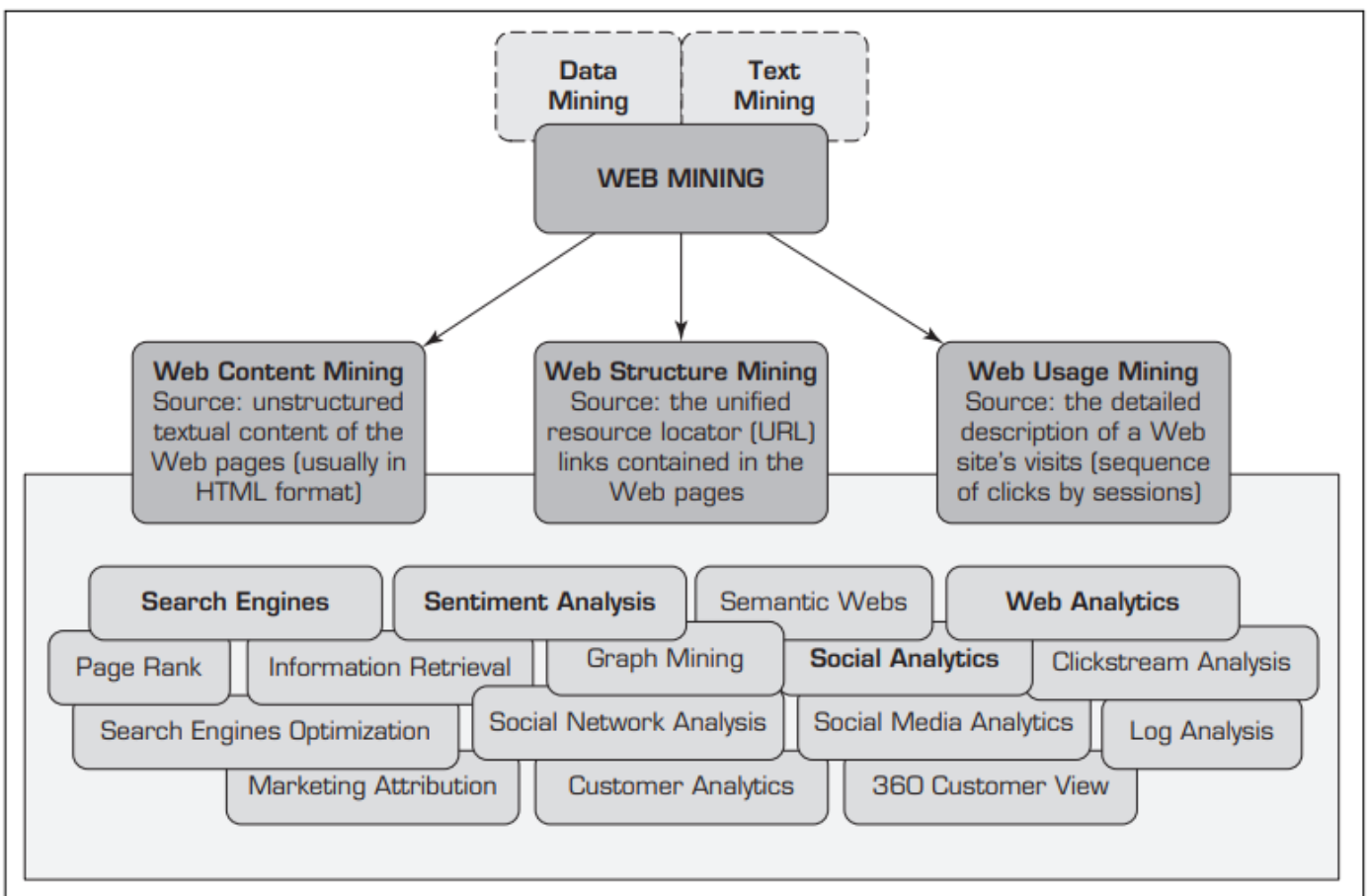


Figure: A Simple Taxonomy of Web Mining

Web Content and Web Structure Min

Web content mining refers to the extraction of useful information from Web pages. The documents may be extracted in some machine-readable format so that automated techniques can extract some information from these Web pages. Web crawlers (also called spiders) are used to read through the content of a Web site automatically. The information gathered may include document characteristics similar to what are used in text mining, but it may also include additional concepts, such as the document hierarchy. Such an automated (or semiautomated) process of collecting and mining Web content can be used for competitive intelligence (collecting intelligence about competitors' products, services, and customers).

It can also be used for information/news/opinion collection and summarization, sentiment analysis, automated data collection, and structuring for predictive modeling. As an illustrative example to using Web content mining as an automated data collection tool, consider the following. For more than 10 years, two of the three authors of this book (Drs. Sharda and Delen) have been developing models to predict the financial success of Hollywood movies before their theatrical release.

The data that they use for training the models come from several Web sites, each of which has a different hierarchical page structure. Collecting a large set of variables on thousands of movies (from the past several years) from these Web sites is a time-demanding, error-prone process. Therefore, they use Web content mining and spiders as an enabling technology to automatically collect, verify, validate (if the specific data item is available on more than one Web site, then the values are validated against each other and anomalies are captured and recorded), and store these values in a relational database. That way, they ensure the quality of the data while saving valuable time (days or weeks) in the process.

In addition to text, Web pages also contain hyperlinks pointing one page to another. Hyperlinks contain a significant amount of hidden human annotation that can potentially help to automatically infer the notion of centrality or authority. When a Web page developer includes a link pointing to another Web page, this may be regarded as the developer's endorsement of the other page. The collective endorsement of a given page by different developers on the Web may indicate the importance of the page and may naturally lead to the discovery of authoritative Web pages (Miller, 2005). Therefore, the vast amount of Web linkage information provides a rich collection of information about the relevance, quality, and structure of the Web's contents, and thus is a rich source for Web mining.

Web content mining can also be used to enhance the results produced by search engines. In fact, search is perhaps the most prevailing application of Web content mining and Web structure mining. A search on the Web to obtain information on a specific topic (presented as a collection of keywords or a sentence) usually returns a few relevant, high-quality Web pages and a larger number of unusable Web pages. Use of a relevance index based on keywords and authoritative pages (or some measure of it) will improve the search results and ranking of relevant pages. The idea of authority (or authoritative pages) stems from earlier information retrieval work using citations among journal articles to evaluate the impact of research papers (Miller, 2005).

Though that was the origin of the idea, there are significant differences between the citations in research articles and hyperlinks on Web pages. First, not every hyperlink represents an endorsement (some links are created for navigation purposes and some are for paid advertisement). While this is true, if the majority of the hyperlinks are of the endorsement type, then the collective opinion will still prevail. Second, for commercial and competitive interests, one authority will rarely have its Web page point to rival authorities in the same domain. For example, Microsoft may prefer not to include links on its Web pages to Apple's Web sites, because this may be regarded as endorsement of its competitor's authority. Third, authoritative pages are seldom particularly

descriptive. For example, the main Web page of Yahoo! may not contain the explicit self-description that it is in fact a Web search engine.

The structure of Web hyperlinks has led to another important category of Web pages called a hub. A hub is one or more Web pages that provide a collection of links to authoritative pages. Hub pages may not be prominent and only a few links may point to them; however, they provide links to a collection of prominent sites on a specific topic of interest. A hub could be a list of recommended links on an individual's homepage, recommended reference sites on a course Web page, or a professionally assembled resource list on a specific topic. Hub pages play the role of implicitly conferring the authorities on a narrow field. In essence, a close symbiotic relationship exists between good hubs and authoritative pages; a good hub is good because it points to many good authorities, and a good authority is good because it is being pointed to by many good hubs. Such relationships between hubs and authorities make it possible to automatically retrieve high-quality content from the Web.

The most popular publicly known and referenced algorithm used to calculate hubs and authorities is **hyperlink-induced topic search (HITS)**. It was originally developed by Kleinberg (1999) and has since been improved on by many researchers. HITS is a link analysis algorithm that rates Web pages using the hyperlink information contained within them. In the context of Web search, the HITS algorithm collects a base document set for a specific query. It then recursively calculates the hub and authority values for each document. To gather the base document set, a root set that matches the query is fetched from a search engine. For each document retrieved, a set of documents that points to the original document and another set of documents that is pointed to by the original document are added to the set as the original document's neighborhood. A recursive process of document identification and link analysis continues until the hub and authority values converge. These values are then used to index and prioritize the document collection generated for a specific query.

Web structure mining is the process of extracting useful information from the links embedded in Web documents. It is used to identify authoritative pages and hubs, which are the cornerstones of the contemporary page-rank algorithms that are central to popular search engines such as Google and Yahoo!. Just as links going to a Web page may indicate a site's popularity (or authority), links within the Web page (or the complete Web site) may indicate the depth of coverage of a specific topic. Analysis of links is very important in understanding the interrelationships among large numbers of Web pages, leading to a better understanding of a specific Web community, clan, or clique. describes a project that used both Web content mining and Web structure mining to better understand how U.S. extremist groups are connected.

Search Engines

In this day and age, there is no denying the importance of Internet search engines. As the size and complexity of the World Wide Web increases, finding what you want is becoming a complex and laborious process. People use search engines for a variety of reasons. We use them to learn about a product or a service before committing to buy (including who else is selling it, what the prices are at different locations/sellers, the common issues people are discussing about it, how satisfied previous buyers are, what other products or services might be better, etc.) and search for places to go, people to meet, and things to do. In a sense, search engines have become the centerpiece of most Internet-based transactions and other activities. The incredible success and popularity of Google, the most popular search engine company, is a good testament to this claim.

. What is somewhat a mystery to many is how a search engine actually does what it is meant to do. In simplest terms, a **search engine** is a software program that searches for documents (Internet sites or files) based on the keywords (individual words, multi-word terms, or a complete sentence) that users have provided that have to do with the subject of their inquiry. Search

engines are the workhorses of the Internet, responding to billions of queries in hundreds of different languages every day.

Technically speaking, search engine is the popular term for information retrieval system. Although Web search engines are the most popular, search engines are often used in a context other than the Web, such as desktop search engines or document search engines. As you will see in this section, many of the concepts and techniques that we covered in the text analytics and text mining previous chapter also apply here. The overall goal of a search engine is to return one or more documents/pages (if more than one documents/ pages applies, a rank-order list is often provided) that best match the user's query. The two metrics that are often used to evaluate search engines are effectiveness (or quality—finding the right documents/pages) and efficiency (or speed—returning a response quickly). These two metrics tend to work in reverse direction; improving one tends to worsen the other. Often, based on user expectation, search engines focus on one at the expense of the other. Better search engines are the ones that excel in both at the same time. Because search engines not only search but, in fact, find and return the documents/pages, perhaps a more appropriate name for them would be “finding engines.

Anatomy of a Search Engine

Now let us dissect a search engine and look inside it. At the highest level, a search engine system is composed of two main cycles: a development cycle and a responding cycle (see the structure of a typical Internet search engine in below Figure). While one is interfacing with the World Wide Web, the other is interfacing with the user. One can think of the development cycle as a production process (manufacturing and inventorying documents/ pages) and the responding cycle as a retailing process (providing customers/users with what they want). In the following section these two cycles are explained in more detail.

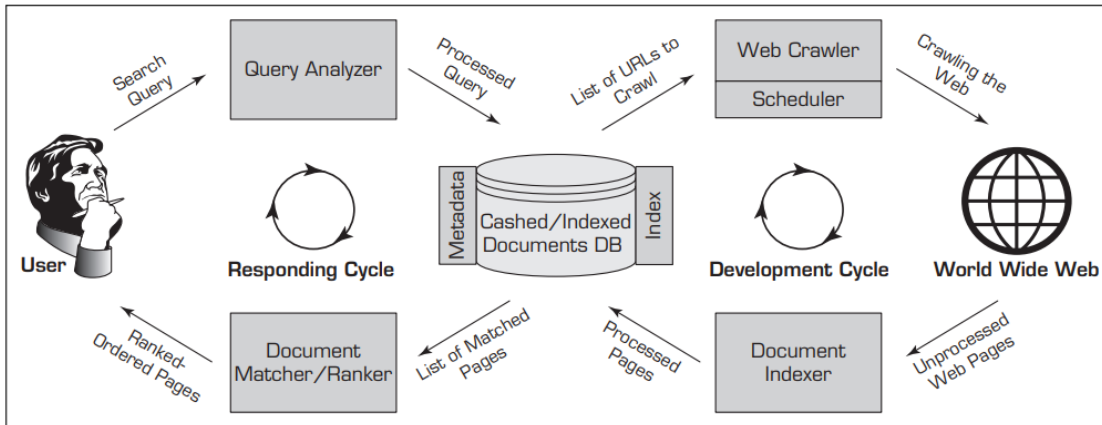


Figure: Structure of a Typical Internet Search Engine

1. Development Cycle

The two main components of the development cycle are the Web crawler and document indexer. The purpose of this cycle is to create a huge database of documents/pages organized and indexed based on their content and information value. The reason for developing such a repository of documents/pages is quite obvious: Due to its sheer size and complexity, searching the Web to find pages in response to a user query is not practical (or feasible within a reasonable time frame); therefore, search engines “cache the Web” into their database, and uses the cached version of the Web for searching and finding. Once created, this database allows search engines to rapidly and accurately respond to user queries.

Web Crawler

A Web crawler (also called a spider or a Web spider) is a piece of software that systematically browses (crawls through) the World Wide Web for the purpose of finding and fetching Web pages. Often Web crawlers copy all the pages they visit for later processing by other functions of a search engine.

A Web crawler starts with a list of URLs to visit, which are listed in the scheduler and often are called the seeds. These URLs may come from submissions made by Webmasters or, more often, they come from the internal hyperlinks of previously crawled documents/pages. As the crawler visits these URLs, it identifies all the hyperlinks in the page and adds them to the list of URLs to visit (i.e., the scheduler). URLs in the scheduler are recursively visited according to a set of policies determined by the specific search

engine. Because there are large volumes of Web pages, the crawler can only download a limited number of them within a given time; therefore, it may need to prioritize its downloads.

Document Indexer

As the documents are found and fetched by the crawler, they are stored in a temporary staging area for the document indexer to grab and process. The document indexer is responsible for processing the documents (Web pages or document files) and placing them into the document database. In order to convert the documents/pages into the desired, easily searchable format, the document indexer performs the following tasks.

Step 1: Preprocessing the Documents Because the documents fetched by the crawler may all be in different formats, for the ease of processing them further, in this step they all are converted to some type of standard representation. For instance, different content types (text, hyperlink, image, etc.) may be separated from each other, formatted (if necessary), and stored in a place for further processing.

Step 2: Parsing the Documents This step is essentially the application of text mining (i.e., computational linguistic, natural language processing) tools and techniques to a collection of documents/pages. In this step, first the standardized documents are parsed into its components to identify index-worthy words/terms. Then, using a set of rules, the words/terms are indexed. More specifically, using tokenization rules, the words/terms/entities are extracted from the sentences in these documents.

Using proper lexicons, the spelling errors and other anomalies in these words/terms are corrected. Not all the terms are discriminators. The nondiscriminating words/terms (also known as stop words) are eliminated from the list of index-worthy words/terms. Because the same word/term can be in many different forms, stemming is applied to reduce the words/terms to their root forms. Again, using lexicons and other language-specific resources (e.g., WordNet), synonyms and homonyms are identified and the word/term collection is processed before moving into the indexing phase.

Step 3: Creating the Term-by-Document Matrix

In this step, the relationships between the words/terms and documents/pages are identified. The weight can be as simple as assigning 1 for presence or 0 for absence of the word/term in the document/page. Usually more sophisticated weight schemas are used. For instance, as opposed to binary, one may choose to assign frequency of occurrence (number of times the same word/term is found in a document) as a weight. Text mining research and practice have clearly indicated that the best weighting may come from the use of

term-frequency divided by inverse-document-frequency (TF/IDF). This algorithm measures the frequency of occurrence of each word/term within a document, and then compares that frequency against the frequency of occurrence in the document collection. As we all know, not all high-frequency words/term are good document discriminators; and a good document discriminator in a domain may not be one in another domain. Once the weighing schema is determined, the weights are calculated and the term-by-document index file is created.

2. Response Cycle

The two main components of the responding cycle are the query analyzer and the document matcher/ranker.

Query Analyzer

The query analyzer is responsible for receiving a search request from the user (via the search engine's Web server interface) and converting it into a standardized data structure, so that it can be easily queried/matched against the entries in the document database. How the query analyzer does what it is supposed to do is quite similar to what the document indexer does (as we have just explained). The query analyzer parses the search string into individual words/terms using a series of tasks that include tokenization, removal of stop words, stemming, and word/term disambiguation (identification of spelling errors, synonyms, and homonyms).

The close similarity between the query analyzer and document indexer is not coincidental. In fact, it is quite logical, because both are working off of the document database; one is putting in documents/pages using a specific index structures, and the other is converting a query string into the same structure so that it can be used to quickly locate most relevant documents/pages.

Document Matcher/Ranker

This is where the structured query data is matched against the document database to find the most relevant documents/pages and also rank them in the order of relevance/ importance. The proficiency of this step is perhaps the most important component when different search engines are compared to one another. Every search engine has its own (often proprietary) algorithm that it uses to carry out this important step.

The early search engines used a simple keyword match against the document database and returned a list of ordered documents/pages, where the determinant of the order was a function that used the number of words/terms matched between the query and the document along with the weights of those words/terms. The quality and the usefulness of the search results were not all that good. Then, in 1997, the creators of Google came up with a new algorithm, called PageRank. As the name implies, PageRank is an algorithmic way to rank-order documents/pages based on their relevance and value/importance. Technology Insights provides a high-level description of this patented algorithm. Even though PageRank is an innovative way to rank documents/pages, it is an augmentation to the process of retrieving relevant documents from the database and ranking them based on the weights of the words/terms. Google does all of these collectively and more to come up with the most relevant list of documents/pages for a given search request. Once an ordered list of documents/pages is created, it is pushed back to the user in an easily digestible format. At this point, users may choose to click on any of the documents in the list, and it may not be the one at the top. If they click on a document/page link that is not at the top of the list, then can we assume that the search engine did not do a good job ranking them? Perhaps, yes. Leading search engines like Google monitor the performance of their search results by capturing, recording, and analyzing postdelivery user actions and experiences. These analyses often lead to more and more rules to further refine the ranking of the documents/pages so that the links at the top are more preferable to the end users.

PageRank Algorithm

PageRank is a link analysis algorithm—named after Larry Page, one of the two inventors of Google, which started as a research project at Stanford University in 1996—used by the Google Web search engine. PageRank assigns a numerical weight to each element of a hyperlinked set of documents, such as the ones found on the World Wide Web, with the purpose of measuring its relative importance within a given collection.

It is believed that PageRank has been influenced by citation analysis, where citations in scholarly works are examined to discover relationships among researchers and their research topics. The applications of citation analysis ranges from identification of prominent experts in a given field of study to providing invaluable information for a transparent review of academic achievements, which can be used for merit review, tenure, and promotion decisions. The PageRank algorithm aims to do the same thing: identifying reputable/important/valuable documents/pages that are highly regarded by other documents/pages. A graphical illustration of PageRank is shown in below figure.

How Does PageRank Work?

Computationally speaking, PageRank extends the citation analysis idea by not counting links from all pages equally and by normalizing by the number of links on a page. PageRank is defined as follows:

Assume page A has pages P1 through Pn pointing to it (with hyperlinks, which is similar to citations in citation analysis). The parameter d is a damping/smoothing factor that can assume values between 0 and 1. Also C(A) is defined as the number of links going out of page A. The simple formula for the PageRank for page A can be written as follows:

$$PageRank(A) = (1 - d) + d \sum_{i=1}^n \frac{PageRank(P_i)}{C(P_i)}$$

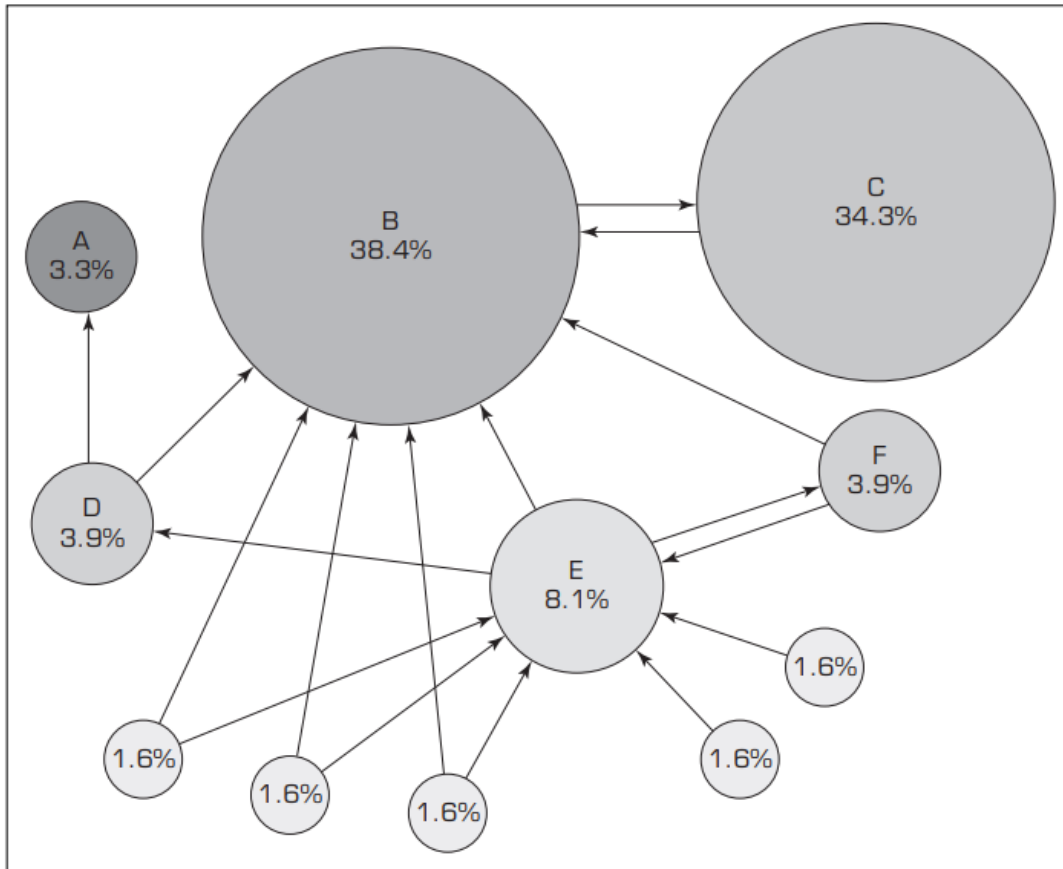


Figure : A Graphical Example for the PageRank Algorithm

Note that the PageRanks form a probability distribution over Web pages, so the sum of all Web pages' PageRanks will be 1. PageRank(A) can be calculated using a simple iterative algorithm and corresponds to the principal eigenvector of the normalized link matrix of the Web. The algorithm is so computationally efficient that a PageRank for 26 million Web pages can be computed in a few hours on a medium-size workstation (Brin and Page, 2012). Of course, there are more details to the actual calculation of PageRank in Google. Most of those details are either not publicly available or are beyond the scope of this simple explanation.

Justification of the Formulation

PageRank can be thought of as a model of user behavior. It assumes there is a random surfer who is given a Web page at random and keeps clicking on

hyperlinks, never hitting back but eventually getting bored and starting on another random page. The probability that the random surfer visits a page is its PageRank. And, the d damping factor is the probability at each page the random surfer will get bored and request another random page. One important variation is to only add the damping factor d to a single page, or a group of pages. This allows for personalization and can make it nearly impossible to deliberately mislead the system in order to get a higher ranking.

Another intuitive justification is that a page can have a high PageRank if there are many pages that point to it, or if there are some pages that point to it and have a high PageRank. Intuitively, pages that are well cited from many places around the Web are worth looking at. Also, pages that have perhaps only one citation from something like the Yahoo! homepage are also generally worth looking at. If a page was not high quality, or was a broken link, it is quite likely that Yahoo!'s homepage would not link to it. The formulation of PageRank handles both of these cases and everything in between by recursively propagating weights through the link structure of the Web.

How Does Google Do It?

Even though complex low-level computational details are trade secrets and are not known to the public, the high-level structure of the Google search system is wellknown and quite simple. From the infrastructure standpoint, the Google search system runs on a distributed network of tens of thousands of computers/servers and can, therefore, carry out its heavy workload effectively and efficiently using sophisticated parallel processing algorithms (a method of computation in which many calculations can be distributed to many servers and performed simultaneously, significantly speeding up data processing). At the highest level, the Google search system has three distinct parts (**googleguide.com**):

1. Googlebot, a Web crawler that roams the Internet to find and fetch Web pages
2. The indexer, which sorts every word on every page and stores the resulting index of words in a huge database

3. The query processor, which compares your search query to the index and recommends the documents that it considers most relevant

1. Googlebot Googlebot is Google's Web crawling robot, which finds and retrieves pages on the Web and hands them off to the Google indexer. It's easy to imagine Googlebot as a little spider scurrying across the strands of cyberspace, but in reality Googlebot doesn't traverse the Web at all. It functions, much like your Web browser, by sending a request to a Web server for a Web page, downloading the entire page, and then handing it off to Google's indexer. Googlebot consists of many computers requesting and fetching pages much more quickly than you can with your Web browser. In fact, Googlebot can request thousands of different pages simultaneously. To avoid overwhelming Web servers, or crowding out requests from human users, Googlebot deliberately makes requests of each individual Web server more slowly than it's capable of doing.

When Googlebot fetches a page, it removes all the links appearing on the page and adds them to a queue for subsequent crawling. Googlebot tends to encounter little spam because most Web authors link only to what they believe are high-quality pages. By harvesting links from every page it encounters, Googlebot can quickly build a list of links that can cover broad reaches of the Web. This technique, known as deep crawling, also allows Googlebot to probe deep within individual sites. Because of their massive scale, deep crawls can reach almost every page in the Web. To keep the index current, Google continuously recrawls popular frequently changing Web pages at a rate roughly proportional to how often the pages change. Such crawls keep an index current and are known as fresh crawls. Newspaper pages are downloaded daily; pages with stock quotes are downloaded much more frequently. Of course, fresh crawls return fewer pages than the deep crawl. The combination of the two types of crawls allows Google to both make efficient use of its resources and keep its index reasonably current.

2. Google Indexer Googlebot gives the indexer the full text of the pages it finds. These pages are stored in Google's index database. This index is sorted

alphabetically by search term, with each index entry storing a list of documents in which the term appears and the location within the text where it occurs. This data structure allows rapid access to documents that contain user query terms. To improve search performance, Google ignores common words, called stop words (such as the, is, on, or, of, a, an, as well as certain single digits and single letters). Stop words are so common that they do little to narrow a search, and therefore they can safely be discarded. The indexer also ignores some punctuation and multiple spaces, as well as converting all letters to lowercase, to improve Google's performance.

3. Google Query Processor The query processor has several parts, including the user interface (search box), the "engine" that evaluates queries and matches them to relevant documents, and the results formatter.

Google uses a proprietary algorithm, called PageRank, to calculate the relative rank order of a given collection of Web pages. PageRank is Google's system for ranking Web pages. A page with a higher PageRank is deemed more important and is more likely to be listed above a page with a lower PageRank. Google considers over a hundred factors in computing a PageRank and determining which documents are most relevant to a query, including the popularity of the page, the position and size of the search terms within the page, and the proximity of the search terms to one another on the page.

Google also applies machine-learning techniques to improve its performance automatically by learning relationships and associations within the stored data. For example, the spelling-correcting system uses such techniques to figure out likely alternative spellings. Google closely guards the formulas it uses to calculate relevance; they're tweaked to improve quality and performance, and to outwit the latest devious techniques used by spammers.

Indexing the full text of the Web allows Google to go beyond simply matching single search terms. Google gives more priority to pages that have search terms near each other and in the same order as the query. Google can also match multi-word phrases and sentences. Because Google indexes HTML

code in addition to the text on the page, users can restrict searches on the basis of where query words appear (e.g., in the title, in the URL, in the body, and in links to the page, options offered by Google's Advanced Search Form and Using Search Operators).

Understanding the internals of popular search engines helps companies, who rely on search engine traffic, better design their e-commerce sites to improve their chances of getting indexed and highly ranked by search providers.

Search Engine Optimization

Search engine optimization (SEO) is the intentional activity of affecting the visibility of an e-commerce site or a Web site in a search engine's natural (unpaid or organic) search results. In general, the higher ranked on the search results page, and more frequently a site appears in the search results list, the more visitors it will receive from the search engine's users. As an Internet marketing strategy, SEO considers how search engines work, what people search for, the actual search terms or keywords typed into search engines, and which search engines are preferred by their targeted audience. Optimizing a Web site may involve editing its content, HTML, and associated coding to both increase its relevance to specific keywords and to remove barriers to the indexing activities of search engines. Promoting a site to increase the number of backlinks, or inbound links, is another SEO tactic.

In the early days, in order to be indexed, all Webmasters needed to do was to submit the address of a page, or URL, to the various engines, which would then send a "spider" to "crawl" that page, extract links to other pages from it, and return information found on the page to the server for indexing. The process, as explained before, involves a search engine spider downloading a page and storing it on the search engine's own server, where a second program, known as an indexer, extracts various information about the page, such as the words it contains and where these are located, as well as any

weight for specific words, and all links the page contains, which are then placed into a scheduler for crawling at a later date. Nowadays search engines are no longer relying on Webmasters submitting URLs (even though they still can); instead, they are proactively and continuously crawling the Web, and finding, fetching, and indexing everything about it.

Being indexed by search engines like Google, Bing, and Yahoo! is not good enough for businesses. Getting ranked on the most widely used search engines (for a list of most widely used search engines) and getting ranked higher than your competitors are what make the difference. A variety of methods can increase the ranking of a Web page within the search results. Cross-linking between pages of the same Web site to provide more links to the most important pages may improve its visibility. Writing content that includes frequently searched keyword phrases, so as to be relevant to a wide variety of search queries, will tend to increase traffic. Updating content so as to keep search engines crawling back frequently can give additional weight to a site. Adding relevant keywords to a Web page's metadata, including the title tag and metadescription, will tend to improve the relevancy of a site's search listings, thus increasing traffic. URL normalization of Web pages so that they are accessible via multiple URLs and using canonical link element and redirects can help make sure links to different versions of the URL all count toward the page's link popularity score.

Methods for Search Engine Optimization

In general, SEO techniques can be classified into two broad categories: techniques that search engines recommend as part of good site design, and those techniques of which search engines do not approve. The search engines attempt to minimize the effect of the latter, which is often called spamdexing (also known as search spam, search engine spam, or search engine poisoning). Industry commentators have classified these methods, and the practitioners who employ them, as either white-hat SEO or black-hat SEO (Goodman, 2005). White hats tend to produce results that last a long time,

whereas black hats anticipate that their sites may eventually be banned either temporarily or permanently once the search engines discover what they are doing.

Here are the 15 most popular search engines as derived from eBizMBA Rank (ebizmba.com/articles/search-engines), which is a constantly updated average of each Web site's Alexa Global Traffic Rank, and U.S. Traffic Rank from both Compete and Quantcast.

Rank	Name	Estimated Unique Monthly Visitors
1	Google	900,000,000
2	Bing	165,000,000
3	Yahoo! Search	160,000,000
4	Ask	125,000,000
5	AOL Search	33,000,000
6	MyWebSearch	19,000,000
7	blekko	9,000,000
8	Lycos	4,300,000
9	Dogpile	2,900,000
10	WebCrawler	2,700,000
11	Info	2,600,000
12	Infospace	2,000,000
13	Search	1,450,000
14	Excite	1,150,000
15	GoodSearch	1,000,000

An SEO technique is considered white hat if it conforms to the search engines' guidelines and involves no deception. Because search engine guidelines are not written as a series of rules or commandments, this is an important distinction to note. White-hat SEO is not just about following guidelines, but about ensuring that the content a search engine indexes and subsequently ranks is the same content a user will see. White-hat advice is generally summed up as creating content for users, not for search engines, and then making that content easily accessible to the spiders, rather than attempting to trick the algorithm from its intended purpose. White-hat SEO is in many ways similar to Web development that promotes accessibility, although the two are not identical.

Black-hat SEO attempts to improve rankings in ways that are disapproved by the search engines, or involve deception. One black-hat technique uses text that is hidden, either as text colored similar to the background, in an invisible div, or positioned off-screen. Another method gives a different page depending

on whether the page is being requested by a human visitor or a search engine, a technique known as cloaking. Search engines may penalize sites they discover using black-hat methods, either by reducing their rankings or eliminating their listings from their databases altogether. Such penalties can be applied either automatically by the search engines' algorithms, or by a manual site review. One example was the February 2006 Google removal of both BMW Germany and Ricoh Germany for use of unapproved practices (Cutts, 2006). Both companies, however, quickly apologized, fixed their practices, and were restored to Google's list.

For some businesses SEO may generate significant return on investment. However, one should keep in mind that search engines are not paid for organic search traffic, their algorithms change constantly, and there are no guarantees of continued referrals. Due to this lack of certainty and stability, a business that relies heavily on search engine traffic can suffer major losses if the search engine decides to change its algorithms and stop sending visitors. According to Google's CEO, Eric Schmidt, in 2010, Google made over 500 algorithm changes—almost 1.5 per day. Because of the difficulty in keeping up with changing search engine rules, companies that rely on search traffic practice one or more of the following: (1) Hire a company that specializes in search engine optimization (there seem to be an abundant number of those nowadays) to continuously improve your site's appeal to changing practices of the search engines; (2) pay the search engine providers to be listed on the paid sponsors sections; and (3) consider liberating yourself from dependence on search engine traffic.

Either originating from a search engine (organically or otherwise) or coming from other sites and places, what is most important for an e-commerce site is to maximize the likelihood of customer transactions. Having a lot of visitors without sales is not what a typical e-commerce site is built for.

Web Usage Mining (Web Analytics)

Web usage mining (also called **Web analytics**) is the extraction of useful information from data generated through Web page visits and transactions. Masand et al. (2002) state that at least three types of data are generated through Web page visits:

1. Automatically generated data stored in server access logs, referrer logs, agent logs, and client-side cookies
2. User profiles
3. Metadata, such as page attributes, content attributes, and usage data.

Analysis of the information collected by Web servers can help us better understand user behavior. Analysis of this data is often called clickstream analysis. By using the data and text mining techniques, a company might be able to discern interesting patterns from the clickstreams. For example, it might learn that 60 percent of visitors who searched for “hotels in Maui” had searched earlier for “airfares to Maui.” Such information could be useful in determining where to place online advertisements. Clickstream analysis might also be useful for knowing when visitors access a site. For example, if a company knew that 70 percent of software downloads from its Web site occurred between 7 and 11 p.m., it could plan for better customer support and network bandwidth during those hours. Figure 8.4 shows the process of extracting knowledge from clickstream data and how the generated knowledge is used to improve the process, improve the Web site, and, most important, increase the customer value.

Web mining has wide a range of business applications. For instance, Nasraoui (2006) listed the following six most common applications:

1. Determine the lifetime value of clients.
2. Design cross-marketing strategies across products.
3. Evaluate promotional campaigns.
4. Target electronic ads and coupons at user groups based on user access patterns.
5. Predict user behavior based on previously learned rules and users' profiles.
6. Present dynamic information to users based on their interests and profiles.

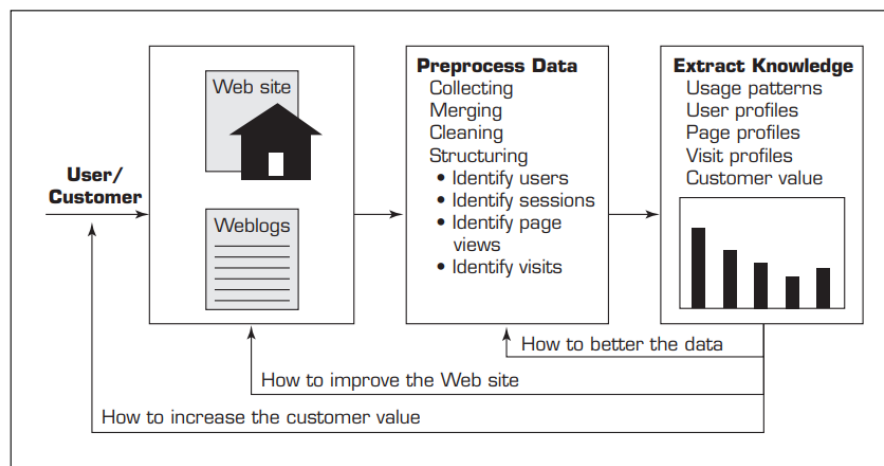


Figure: Extraction of Knowledge from Web Usage Data.

Amazon.com provides an excellent example of how Web usage history can be leveraged dynamically. A registered user who revisits Amazon.com is greeted by name. This is a simple task that involves recognizing the user by reading a cookie (i.e., a small text file written by a Web site on the visitor's computer). Amazon.com also presents the user with a choice of products in a personalized store, based on previous purchases and an association analysis of similar users. It also makes special "Gold Box" offers that are good for a short amount of time. All these recommendations involve a detailed analysis of the visitor as well as the user's peer group developed through the use of clustering, sequence pattern discovery, association, and other data and text mining techniques.

Web Analytics Technologies

There are numerous tools and technologies for Web analytics in the marketplace. Because of their power to measure, collect, and analyze Internet data to better understand and optimize Web usage, the popularity of Web analytics tools is increasing. Web analytics holds the promise to revolutionize how business is done on the Web. Web analytics is not just a tool for measuring Web traffic; it can also be used as a tool for e-business and market research, and to assess and improve the effectiveness of an e-commerce Web site. Web analytics applications can also help companies measure the results of traditional print or broadcast advertising campaigns. It can help estimate how traffic to a Web site changes after the launch of a new advertising campaign. Web analytics provides information about the number of visitors to a Web site and the number of page views. It helps gauge traffic and popularity trends, which can be used for market research.

There are two main categories of web analytics; off-site and on-site. Off-site Web analytics refers to Web measurement and analysis about you and your products that takes place outside your Web site. It includes the measurement of a Web site's potential audience (prospect or opportunity), share of voice (visibility or word-of-mouth), and buzz (comments or opinions) that is happening on the Internet.

What is more mainstream is on-site Web analytics. Historically, Web analytics has referred to on-site visitor measurement. However, in recent years this has blurred, mainly because vendors are producing tools that span both categories. On-site Web analytics measure a visitors' behavior once they are on your Web site. This includes its drivers and conversions—for example, the degree to which different landing pages are associated with online purchases. On-site Web analytics measure the performance of your Web site in a commercial context. This data collected on the Web site is then compared against key performance indicators for performance, and used to improve a Web site's or marketing campaign's audience response. Even though Google Analytics is the most widely-used onsite Web analytics service, there are others provided by Yahoo! and Microsoft, and newer and better tools are emerging constantly that provide additional layers of information.

For on-site Web analytics, there are two technical ways of collecting the data. The first and more traditional method is the server log file analysis, where the Web server records file requests made by browsers. The second method is page tagging, which uses JavaScript embedded in the site page code to make image requests to a third-party analytics-dedicated server whenever a page is rendered by a Web browser (or when a mouse click occurs). Both collect data that can be processed to produce Web traffic reports. In addition to these two main streams, other data sources may also be added to augment Web site behavior data. These other sources may include e-mail, direct-mail campaign data, sales and lead history, or social media–originated data.

Web Analytics Metrics

Using a variety of data sources, Web analytics programs provide access to a lot of valuable marketing data, which can be leveraged for better insights to grow your business and better document your ROI. The insight and intelligence gained from Web analytics can be used to effectively manage the marketing efforts of an organization and its various products or services. Web analytics programs provide nearly real-time data, which can document your marketing campaign successes or empower you to make timely adjustments to your current marketing strategies.

While Web analytics provides a broad range of metrics, there are four categories of metrics that are generally actionable and can directly impact your business objectives (TWG, 2013). These categories include:

- Web site usability: How were they using my Web site?
- Traffic sources: Where did they come from?
- Visitor profiles: What do my visitors look like?
- Conversion statistics: What does all this mean for the business?

Web Site Usability

Beginning with your Web site, let's take a look at how well it works for your visitors. This is where you can learn how “user-friendly” it really is or whether or not you are providing the right content.

- 1. Page views.**
- 2. Time on site**
- 3. Downloads.**
- 4. Click map.**
- 5. Click paths.**

Traffic Sources

Your Web analytics program is an incredible tool for identifying where your Web traffic originates. Basic categories such as search engines, referral Web sites, and visits from bookmarked pages (i.e., direct) are compiled with little involvement by the marketer. With a little effort, however, you can also identify Web traffic that was generated by your various offline or online advertising campaigns.

- 1. Referral Web sites**
- 2. Search engines**
- 3. Direct**
- 4. Offline campaigns**
- 5. Online campaigns**

Visitor Profiles

One of the ways you can leverage your Web analytics into a really powerful marketing tool is through segmentation. By blending data from different analytics reports, you'll begin to see a variety of user profiles emerge.

- 1. Keywords**
- 2. Content groupings**
- 3. Geography.**
- 4. Time of day**
- 5. Landing page profiles**

Conversion Statistics

Each organization will define a “conversion” according to its specific marketing objectives. Some Web analytics programs use the term “goal” to benchmark certain Web site objectives, whether that be a certain number of visitors to a page, a completed registration form, or an online purchase.

- 1. New visitors.**
- 2. Returning visitors**
- 3. Leads**
- 4. Sales/conversions**
- 5. Abandonment/exit rates**

Web Analytics Maturity Model and Web Analytics Tools

The term “maturity” relates to the degree of proficiency, formality, and optimization of business models, moving “ad hoc” practices to formally defined steps and optimal business processes. A maturity model is a formal depiction of critical dimensions and their competency levels of a business practice. Collectively, these dimensions and levels define the maturity level of an organization in that area of practice. It often describes an evolutionary improvement path from ad hoc, immature practices to disciplined, mature processes with improved quality and efficiency.

A good example of maturity models is the BI Maturity Model developed by The Data Warehouse Institute (TDWI). In the TDWI BI Maturity Model the main purpose was to gauge where organization data warehousing initiatives are at a point in time and where it should go next. It was represented in a six-stage framework (Management Reporting → Spreadmarts → Data Marts → Data Warehouse → Enterprise Data Warehouse → BI Services). Another related example is the simple business analytics maturity model, moving from simple descriptive

measures to predicting future outcomes, to obtaining sophisticated decision systems (i.e., Descriptive Analytics → Predictive Analytics → Prescriptive Analytics).

For Web analytics perhaps the most comprehensive model was proposed by Stéphane Hamel (2009). In this model, Hamel used six dimensions—(1) Management, Governance and Adoption, (2) Objectives Definition, (3) Scoping, (4) The Analytics Team and Expertise, (5) The Continuous Improvement Process and Analysis Methodology, (6) Tools, Technology and Data Integration—and for each dimension he used six levels of proficiency/competence. Figure 8.7 shows Hamel's six dimensions and the respective proficiency levels.

The proficiency/competence levels have different terms/labels for each of the six dimensions, describing specifically what each level means. Essentially, the six levels are indications of analytical maturity ranging from “0—Analytically Impaired” to “5—Analytical Competitor.” A short description of each of the six levels of competencies is given here (Hamel, 2009):

1. **Impaired:** Characterized by the use of out-of-the-box tools and reports; limited resources lacking formal training (hands-on skills) and education (knowledge). Web analytics is used on an ad hoc basis and is of limited value and scope. Some tactical objectives are defined, but results are not well communicated and there are multiple versions of the truth.

2. **Initiated:** Works with metrics to optimize specific areas of the business (such as marketing or the e-commerce catalogue). Resources are still limited, but the process is getting streamlined. Results are communicated to various business stakeholders (often director level). However, Web analytics might be supporting obsolete business processes and, thus, be limited in the ability to push for optimization beyond the online channel. Success is mostly anecdotal.

3. **Operational:** Key performance indicators and dashboards are defined and aligned with strategic business objectives. A multidisciplinary team is in place and uses various sources of information such as competitive data, voice of customer, and social media or mobile analysis. Metrics are exploited and explored through segmentation and multivariate testing. The Internet channel is being optimized; personas are being defined. Results start to appear and be considered at the executive level. Results are centrally driven, but broadly distributed.

4. **Integrated:** Analysts can now correlate online and offline data from various sources to provide a near 360-degree view of the whole value chain. Optimization encompasses complete processes, including back-end and front-end. Online activities are defined from the user perspective and persuasion scenarios are defined. A continuous improvement process and problem-solving methodologies are prevalent. Insight and recommendations reach the CXO level.

5. **Competitor:** This level is characterized by several attributes of companies with a strong analytical culture (Davenport and Harris, 2007):

- a. One or more senior executives strongly advocate fact-based decision making and analytics
- b. Widespread use of not just descriptive statistics, but predictive modeling and complex optimization techniques
- c. Substantial use of analytics across multiple business functions or processes
- d. Movement toward an enterprise-level approach to managing analytical tools, data, and organizational skills and capabilities.

6. **Addicted:** This level matches Davenport's "Analytical Competitor" characteristics: deep strategic insight, continuous improvement, integrated, skilled resources, top management commitment, fact-based culture, continuous testing, learning, and most important: far beyond the boundaries of the online channel.

In Below Figure, one can mark the level of proficiency in each of the six dimensions to create their organization's maturity model (which would look like a spider diagram).

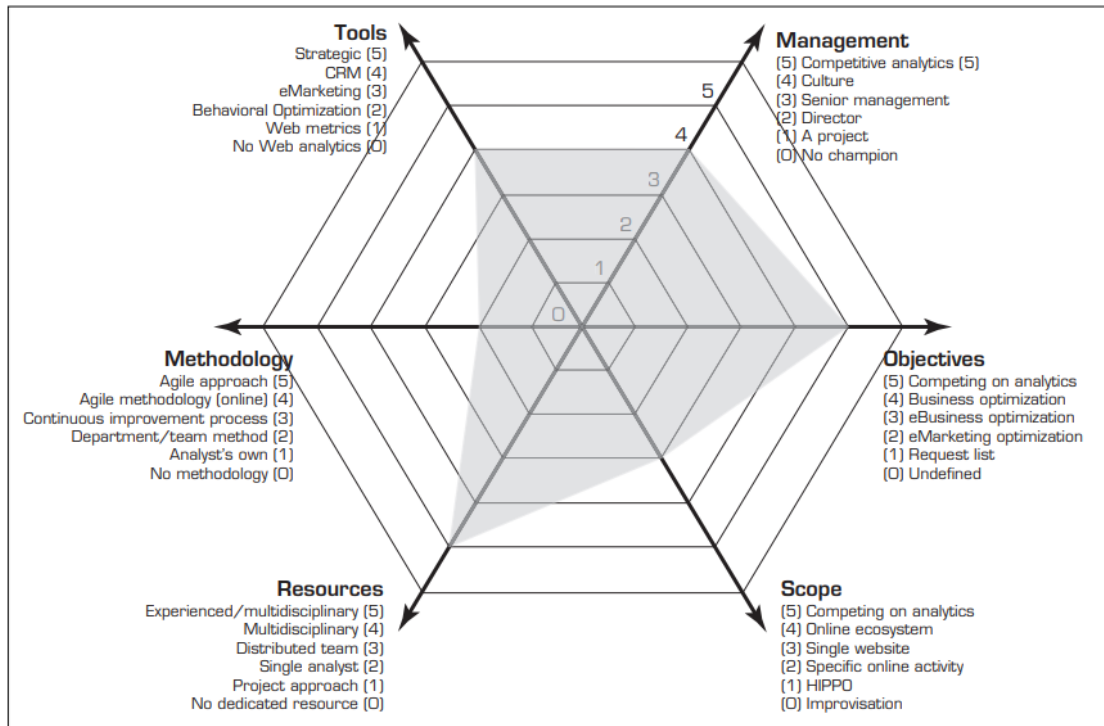


Fig: A Framework for Web Analytics Maturity Model

Such an assessment can help organizations better understand at what dimensions they are lagging behind, and take corrective actions to mitigate it.

Web Analytics Tools

There are plenty of Web analytics applications (downloadable software tools and Webbased/on-demand service platforms) in the market. Companies (large, medium, or small) are creating products and services to grab their fair share from the emerging Web analytics marketplace. What is the most interesting is that many of the most popular Web analytics tools are free—yes, free to download and use for whatever reasons, commercial or nonprofit. The following are among the most popular free (or almost free) Web analytics tools:

GOOGLE WEB ANALYTICS ([GOOGLE.COM/ANALYTICS](https://www.google.com/analytics)) This is a service offered by Google that generates detailed statistics about a Web site's traffic and traffic sources and measures conversions and sales. The product is aimed at marketers as opposed to the Webmasters and technologists from which the industry of Web analytics originally grew. It is the most widely used Web analytics service. Even though the basic service is free of charge, the premium version is available for a fee.

YAHOO! WEB ANALYTICS ([WEB.ANALYTICS.YAHOO.COM](https://web.analytics.yahoo.com)) Yahoo! Web analytics is Yahoo!'s alternative to the dominant Google Analytics. It's an enterprise-level, robust Web-based third-party solution that makes accessing data easy, especially for multiple-user groups. It's got all the things you'd expect from a comprehensive Web analytics tool, such as pretty graphs, custom-designed (and printable) reports, and real-time data tracking.

OPEN WEB ANALYTICS ([OPENWEBANALYTICS.COM](https://openwebanalytics.com)) Open Web Analytics (OWA) is a popular open source Web analytics software that anyone can use to track and analyze how people use Web sites and applications. OWA is licensed under GPL and provides Web site owners and developers with easy ways to add Web analytics to their sites using simple Javascript, PHP, or REST-based APIs. OWA also comes with built-in support for tracking Web sites made with popular content management frameworks such as WordPress and MediaWiki.

PIWIK ([PIWIK.ORG](https://piwik.org)) Piwik is the one of the leading self-hosted, decentralized, open source Web analytics platforms, used by 460,000 Web sites in 150 countries. Piwik was founded by Matthieu Aubry in 2007. Over the last 6 years, more talented and passionate members of the community have joined the team. As is the case in many open source initiatives, they are actively looking for new developers, designers, datavis architects, and sponsors to join them.

FIRESTAT ([FIRESTATS.CC](https://firestats.cc)) FireStats is a simple and straightforward Web analytics application written in PHP/MySQL. It supports numerous platforms and set-ups including C# sites, Django sites, Drupal, Joomla!, WordPress, and several others. FireStats has an intuitive API that assists developers in creating their own custom apps or publishing platform components.

SITE METER ([SITEMETER.COM](https://www.sitemeter.com)) Site Meter is a service that provides counter and tracking information for Web sites. By logging IP addresses and using JavaScript or HTML to track visitor information, Site Meter provides Web site owners with information about their visitors, including how they reached the site, the date and time of their visit, and more.

WOOPRA (WOOPRA.COM) Woopra is a real-time customer analytics service that provides solutions for sales, service, marketing, and product teams. The platform is designed to help organizations optimize the customer life cycle by delivering live, granular behavioral data for individual Web site visitors and customers. It ties this individual-level data to aggregate analytics reports for a full life-cycle view that bridges departmental gaps.

AWSTATS (AWSTATS.ORG) AWStats is an open source Web analytics reporting tool, suitable for analyzing data from Internet services such as Web, streaming media, mail, and FTP servers. AWStats parses and analyzes server log files, producing HTML reports. Data is visually presented within reports by tables and bar graphs. Static reports can be created through a command line interface, and on-demand reporting is supported through a Web browser CGI program.

SNOOP (REINVIGORATE.NET) Snoop is a desktop-based application that runs on the Mac OS X and Windows XP/Vista platforms. It sits nicely on your system status bar/system tray, notifying you with audible sounds whenever something happens. Another outstanding Snoop feature is the Name Tags option, which allows you to “tag” visitors for easier identification. So when Joe over at the accounting department visits your site, you’ll instantly know.

MOCHIBOT (MOCHIBOT.COM) MochiBot is a free Web analytics/tracking tool especially designed for Flash assets. With MochiBot, you can see who’s sharing your Flash content, how many times people view your content, as well as help you track where your Flash content is to prevent piracy and content theft. Installing MochiBot is a breeze; you simply copy a few lines of ActionScript code in the .FLA files you want to monitor.

In addition to these free Web analytics tools, Table 8.1 provides a list of commercially available Web analytics tools.